

		Department of Computer Science & Engineering (CSE) School of Engineering, Technology & Sciences INDEPENDENT UNIVERSITY, BANGLADESH (IUB)																			
Computer Science and Engineering (CSE) Program (Undergraduate)																					
MINI Project Report		CSE 303 Lab		Summer 2026																	
<table border="1"> <tr> <td>Course Code</td> <td>CSE303</td> <td>Course Title</td> <td>Database Management</td> </tr> <tr> <td>Course Type</td> <td>Core</td> <td>Co-offered (if any)</td> <td>CSE303, CSC401, CEN401</td> </tr> <tr> <td>Credit Value</td> <td>4 (T: 3, L: 1)</td> <td>Contact Hours / Week</td> <td>3 hours (theory) 2 hours (Lab)</td> </tr> <tr> <td>Year</td> <td>3rd Year</td> <td>Semester</td> <td>7th Semester</td> </tr> </table>						Course Code	CSE303	Course Title	Database Management	Course Type	Core	Co-offered (if any)	CSE303, CSC401, CEN401	Credit Value	4 (T: 3, L: 1)	Contact Hours / Week	3 hours (theory) 2 hours (Lab)	Year	3 rd Year	Semester	7 th Semester
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Name: Mr. XYZ		ID: 22025214		Sec: 1-(3)																	
CLO4	Apply cutting edge DBMS tools and techniques to implement solutions to a complex database problem with rational justification.		PLO-e: Modern tool usage: Select/apply modern engineering & IT tools (prediction/modeling) appropriately and understand their limitations.																		
Tasks/Topics covered and Skill Measured: - Designing the database (ER-Diagram with detail specifications). - Create and Manage the Database using MySQL. - User Access control. - Application of Database Normalization Techniques. - Scenario based Database Recovery Mechanism. - Application of Hashing and Indexing. - Database Connectivity and front-end design.																					
Acronyms and Legends used: MA – Marks Allotted MO – Marks Obtained																					
MA - 150			MO- Out of 150 - Out of 15 -																		
Project Definition - E-Commerce Database System																					
Designing an E-Commerce Database An online retail company wants to develop an E-Commerce Management System to manage products, categories, customers, orders, suppliers, warehouses, and payments.																					

Products are organized into categories. Each category has category ID, category name, and description. Each product has a unique product ID, product name, brand, price, stock quantity, description, and availability status.

Suppliers provide products. Each supplier has supplier ID, company name, contact person, phone number, email, and address. Customers create accounts on the platform. Each customer has customer ID, full name, email, phone number, address, registration date, and membership level.

Customers place orders containing one or more products. Each order has order ID, order date, order status, shipping address, and total amount. Order details store product quantity, unit price, and subtotal. Products are stored in warehouses. Each warehouse has warehouse ID, warehouse name, location, and storage capacity.

Payments are processed for orders. Each payment contains payment ID, payment date, amount, payment method, and payment status. The system should answer questions such as: best-selling products, pending orders, low-stock products, supplier-wise inventory, and sales revenue during a period.

Task -1	Designing the database	MA - 20	MO --	
Identify and list all entities, attributes for each entity, domain of each attribute, candidate keys and primary key for each entity. Each row should be used for one entity. Add/remove rows as appropriate.				
Entity name	Attribute list	Domain of each attribute	Candidate key(s) for each entity	Primary key for the entity
MA - 2	MA - 2	MA – 3	MA – 2	MA - 1
MO -	MO -	MO -	MO -	MO -
Draw the detailed ER diagram for the identified entity set and associated information. Your design must include: entity sets with all attributes, attribute types and domain, keys, relationships with cardinality, relationship sets (with attributes), weak entity sets, roles in relationships. All appropriate symbols and notations must be used to prepare the diagram.				
MA (Completeness) - 3	MA (Accuracy) - 3	MA (VIVA) - 4		
MO -	MO -	MO -		

Task -2	Create, Manage and Access Control of the Database using MySQL.	MA - 35	MO --
Create all the Tables, Relation between tables with all the attributes (domain), integrity constrains and relations based on the above ER Diagram.			
MA (Completeness) - 2		MA (Accuracy) - 1	MA (VIVA) - 2
MO -		MO -	MO -
Write comprehensive SQL Queries to perform the followings: a. Show customer name, email, and membership level for all customers. b. Display product name, brand, and price for all products. c. Find customers whose names contain "man" and show their email addresses. d. Show order ID, customer name, order date, and total amount for all orders. e. Find products whose price is greater than the average price of all products. f. Display the total number of products available in each category. g. Find categories where the average product price is greater than 5,000. h. Show warehouse name, location, and storage capacity for all warehouses. i. Display customer name and order date for all pending orders. j. Find the total purchase amount made by each customer.			
MA (Completeness) - 5		MA (Accuracy) - 5	MA (VIVA) - 5
MO -		MO -	MO -
Write comprehensive SQL triggers to perform the following:			

a. Automatically check product stock availability before inserting an order detail. If the requested quantity exceeds the available stock, prevent the order. Write a sample INSERT statement showing how the trigger works. b. Recalculate the final payment amount before updating a payment record. The payment amount is calculated as follows: <i>Final_Amount = Total_Amount + Tax - Discount</i> The trigger should automatically update the Final_Amount whenever the payment information is modified. c. Store deleted order information in an order log table. Whenever an order record is deleted, the trigger should save all relevant order details along with the deletion date and time into the log table.																															
MA (Completeness) - 5		MA (Accuracy) - 5																													
MA (VIVA) - 5																															
MO -		MO -																													
MO -		MO -																													
Task -3		Scenario based Database Recovery Mechanism.																													
MA - 18		MO -																													
The e-commerce database backup system uses RAID Level 4 to improve data reliability and ensure continuous access to online store information. In RAID Level 4, data is divided into fixed-size blocks and stored across multiple data disks using block-level striping. A separate dedicated parity disk stores the parity block. The parity block is calculated using the XOR operation on all data blocks. In this system, there are: • 6 data disks: D1, D2, D3, D4, D5, D6 • 1 dedicated parity disk: P • Each disk block stores 4 bits • Parity is calculated using XOR operation The following database blocks are stored in the RAID 4 system: <table><tr><th>Data Block</th><th>Description</th><th>4-bit Binary Value</th><th>Stored in</th></tr><tr><td>Block 1 (D1)</td><td>Product Data</td><td>1010</td><td>D1</td></tr><tr><td>Block 2 (D2)</td><td>Category Data</td><td>1100</td><td>D2</td></tr><tr><td>Block 3 (D3)</td><td>Customer Data</td><td>0111</td><td>D3</td></tr><tr><td>Block 4 (D4)</td><td>Supplier Data</td><td>1001</td><td>D4</td></tr><tr><td>Block 5 (D5)</td><td>Order Data</td><td>0011</td><td>D5</td></tr><tr><td>Block 6 (D6)</td><td>Payment Data</td><td>1110</td><td>D6</td></tr></table> The parity block is stored in the dedicated parity disk P. Now perform the following:				Data Block	Description	4-bit Binary Value	Stored in	Block 1 (D1)	Product Data	1010	D1	Block 2 (D2)	Category Data	1100	D2	Block 3 (D3)	Customer Data	0111	D3	Block 4 (D4)	Supplier Data	1001	D4	Block 5 (D5)	Order Data	0011	D5	Block 6 (D6)	Payment Data	1110	D6
Data Block	Description	4-bit Binary Value	Stored in																												
Block 1 (D1)	Product Data	1010	D1																												
Block 2 (D2)	Category Data	1100	D2																												
Block 3 (D3)	Customer Data	0111	D3																												
Block 4 (D4)	Supplier Data	1001	D4																												
Block 5 (D5)	Order Data	0011	D5																												
Block 6 (D6)	Payment Data	1110	D6																												

- Draw the RAID Level 4 structure.
- Show the detailed calculation of the parity block (P).
- Assume that Data Block D5 (Order Data) fails. Show the detailed calculation using the parity block and the remaining data blocks to recover the lost data.
- Verify that the recovered value of D5 matches the original Order Data block.
- Briefly explain why RAID Level 4 can recover data when a single disk fails.

MA (Completeness) - 6		MA (VIVA) - 12
MO -		MO -

Task -4	Database Normalization.	MA - 22	MO -
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Give an Unnormalized sample database table for the E-commerce Database system.

<u>Product ID</u>	<u>Product Name</u>	<u>Category Name</u>	<u>Supplier Name</u>	<u>Customer ID</u>	<u>Customer Name</u>	<u>Order ID</u>	<u>Products Ordered</u>	<u>Payment Amount</u>
P001	Laptop	Electronics	Tech Ltd	C001	Rahim Ahmed	O001	Mouse, Keyboard	85000
P002	Mobile	Electronics	Smart Corp	C002	Karim Hasan	O002	Charger, Earphone	45000
P003	Printer	Office Equipment	Print Solutions	C003	Nila Sultana	O003	Ink Cartridge, Paper	25000
P004	Tablet	Electronics	Digital World	C004	Arif Khan	O004	Cover, Screen Protector	35000
P005	Office Chair	Furniture	Comfort Furniture	C005	Nusrat Jahan	O005	Cushion, Foot Rest	18000
P006	External Hard Drive	Computer Accessories	Storage Tech	C006	Fahim Hossain	O006	USB Cable, Carry Case	12000

The parity block is stored in the dedicated parity **disk P**.

Now perform the following:

- Apply 1NF to normalize the given table and produce the resultant table.
- Apply 2NF to normalize the table(s) produced in 1NF and present the resultant tables.
- Apply 3NF to normalize the table(s) produced in 2NF and present the resultant tables.

MA (Completeness) - 6	Ma (Correctness) - 6	MA (VIVA) - 10
MO -	MO-	MO -

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Task -5	Hashing and Indexing.	MA - 25	MO -
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Scenario

The E-Commerce Management Database System maintains a **PRODUCT** table containing information about products. The database administrator wants to create an index on **Product_ID** so that product records can be searched efficiently.

The following patient records are inserted into the table in ascending order of **Product_ID**.

PRODUCT Table

Product ID	Product Name
101	Laptop
102	Mobile Phone
103	Printer
104	Keyboard
105	Mouse
106	Monitor
107	Scanner
108	Router
109	SSD
110	Graphics Card
111	Webcam
112	Power Bank
113	Smart Watch

The index key is: **Product_ID**

The insertion sequence is: 110, 104, 101, 113, 107, 109, 102, 111, 105, 108, 103, 112, 106

Assume:

- Every node can contain at most 3 keys.
- The trees must remain balanced.

Now perform the following:

a. Apply the B-Tree algorithm to create a B-Tree for the PRODUCT table. b. Apply the B+ Tree algorithm to create a B+ Tree for the PRODUCT table. c. Demonstrate how the system searches for Product_ID = 108 using the B+ Tree. d. Demonstrate how the system deletes Product_ID = 105 from the B-Tree.		
MA (Completeness) - 6	Ma (Correctness) - 6	MA (VIVA) - 13
MO -	MO-	MO -

Task -6	User Authentication and Database Connectivity.	MA - 30	MO -
Develop a simple E-Commerce Management Web Application with user authentication and database connectivity. The system should provide: User Registration: New users can create an account using a username, email, and password. User Login: Registered users can log in securely using their credentials. Product Management Page: After successful login, users can access a page to manage product records. - The page should support basic CRUD operations: Create: Add a new product record. Read: View all product records. Update: Edit existing product information. Delete: Remove a product record from the database. The application must connect to a database to store user accounts and product information permanently.			
MA (Completeness) - 6		MA (VIVA) - 12	
MO -		MO -	